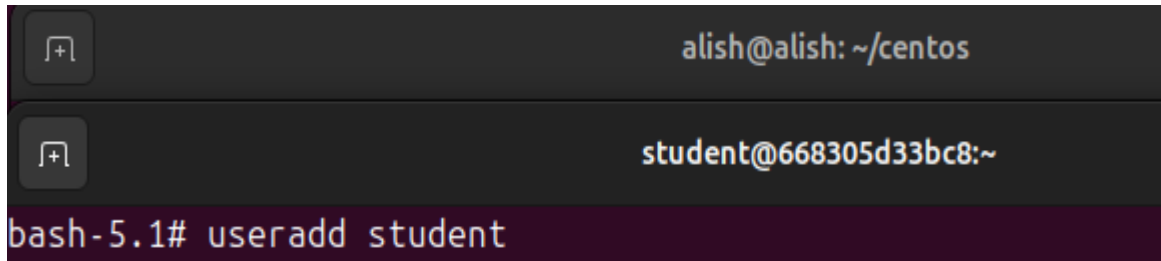


A. Create a user named “student”

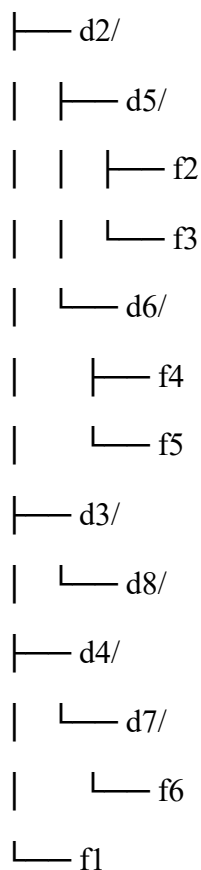
Command : useradd student



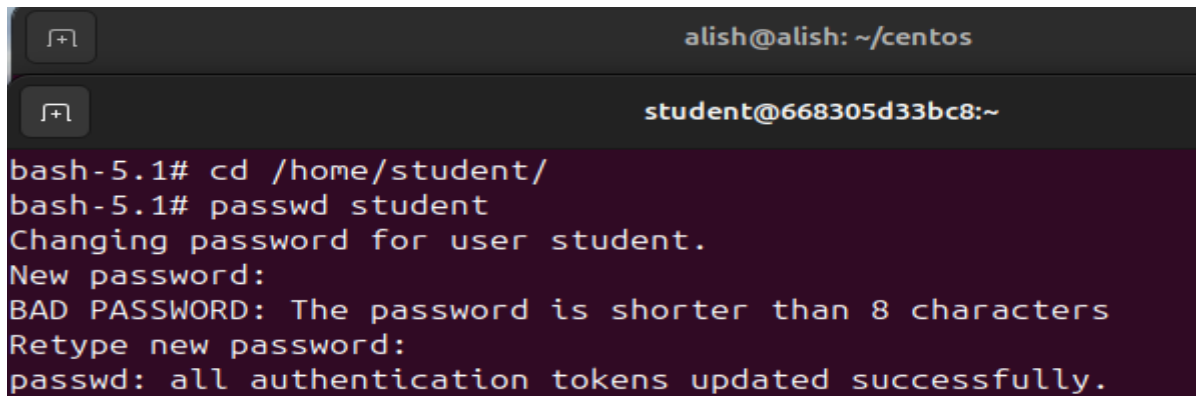
The image shows a terminal window with a dark background. The top part shows the prompt 'alish@alish: ~/centos'. Below that, the prompt changes to 'student@668305d33bc8:~', indicating a successful login as the 'student' user. The bottom part of the terminal shows the command 'bash-5.1# useradd student' being entered.

B. Login from “student” user then create a file and folder according to the following tree structure

d1/



Command: user student
passwd student

A terminal window with a dark background and light-colored text. The window title bar shows 'alish@alish: ~/centos'. The terminal content shows a user switching to 'student' and then using 'passwd' to change the password. The password is rejected as being too short (less than 8 characters) and is then successfully updated.

```
alish@alish: ~/centos
student@668305d33bc8:~
bash-5.1# cd /home/student/
bash-5.1# passwd student
Changing password for user student.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
```

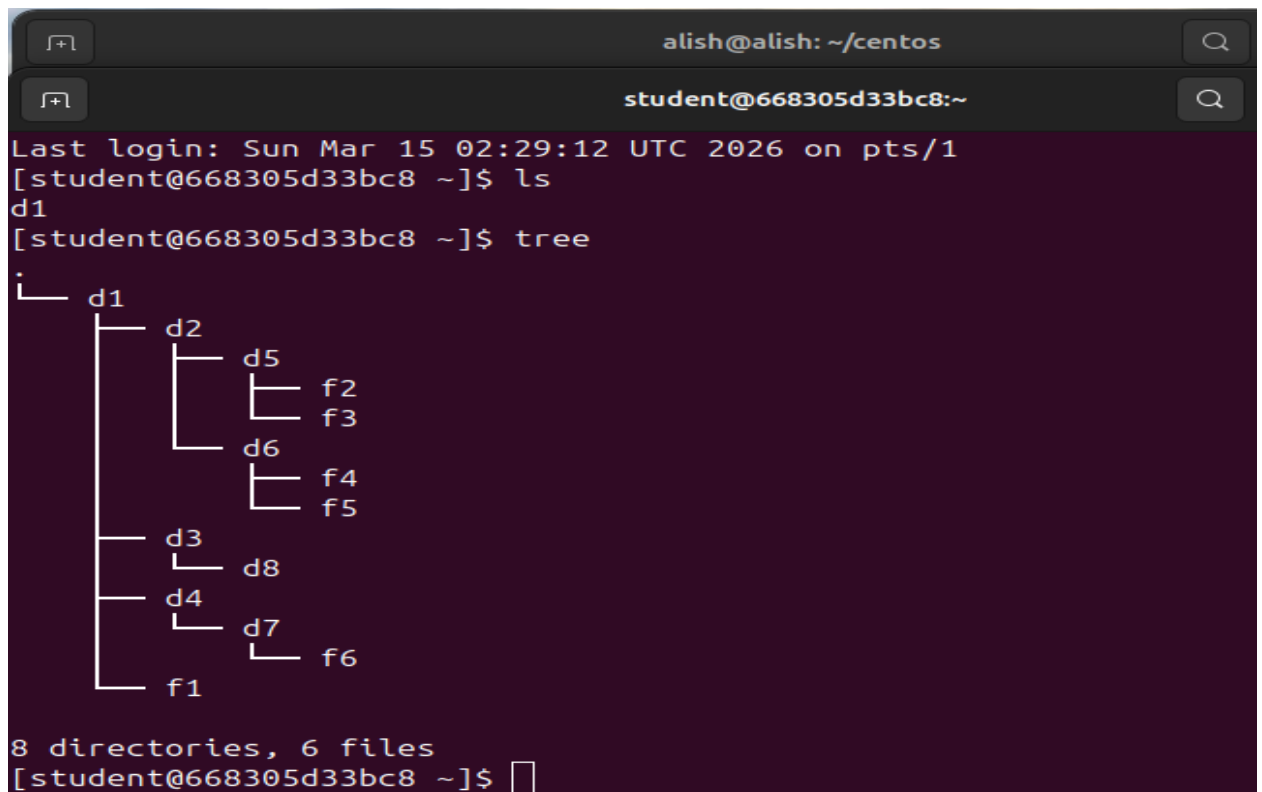
Command:

- mkdir d1
- cd d1/
- mkdir d2 d3 d4
- touch f1
- cd d2
- mkdir d5 d6
- cd d5
- touch f2 f3
- cd ..
- cd d6
- touch f4 f5
- cd ~
- cd d1
- cd d3
- mkdir d8
- cd ..
- cd d4
- mkdir d7
- cd d7
- touch f6
- cd ~

```
alish@alish: ~/centos
student@668305d33bc8:~
[student@668305d33bc8 ~]$ mkdir d1
[student@668305d33bc8 ~]$ ls
d1
[student@668305d33bc8 ~]$ cd d1/
[student@668305d33bc8 d1]$ mkdir d2 d3 d4
[student@668305d33bc8 d1]$ ls
d2 d3 d4
[student@668305d33bc8 d1]$ touch f1
[student@668305d33bc8 d1]$ ls
d2 d3 d4 f1
[student@668305d33bc8 d1]$ cd d2/
[student@668305d33bc8 d2]$ mkdir d5 d6
[student@668305d33bc8 d2]$ ls
d5 d6
[student@668305d33bc8 d2]$ cd d5/
[student@668305d33bc8 d5]$ touch f2 f3
[student@668305d33bc8 d5]$ ls
f2 f3
[student@668305d33bc8 d5]$ cd ..
[student@668305d33bc8 d2]$ ls
d5 d6
[student@668305d33bc8 d2]$ cd d6/
[student@668305d33bc8 d6]$ touch f4 f5
[student@668305d33bc8 d6]$ ls
f4 f5
[student@668305d33bc8 d6]$ cd ..
```

```
alish@alish: ~/centos
student@668305d33bc8:~
[student@668305d33bc8 d6]$ cd ..
[student@668305d33bc8 d2]$ cd ..
[student@668305d33bc8 d1]$ ls
d2 d3 d4 f1
[student@668305d33bc8 d1]$ cd d3/
[student@668305d33bc8 d3]$ mkdir d8
[student@668305d33bc8 d3]$ ls
d8
[student@668305d33bc8 d3]$ cd ..
[student@668305d33bc8 d1]$ cd ..
[student@668305d33bc8 ~]$ ls
d1
[student@668305d33bc8 ~]$ ls
d1
[student@668305d33bc8 ~]$ cd d1/
[student@668305d33bc8 d1]$ ls
d2 d3 d4 f1
[student@668305d33bc8 d1]$ cd d4/
[student@668305d33bc8 d4]$ mkdir d7
[student@668305d33bc8 d4]$ ls
d7
[student@668305d33bc8 d4]$ cd d7
[student@668305d33bc8 d7]$ touch f6
[student@668305d33bc8 d7]$ ls
f6
```

Command: tree

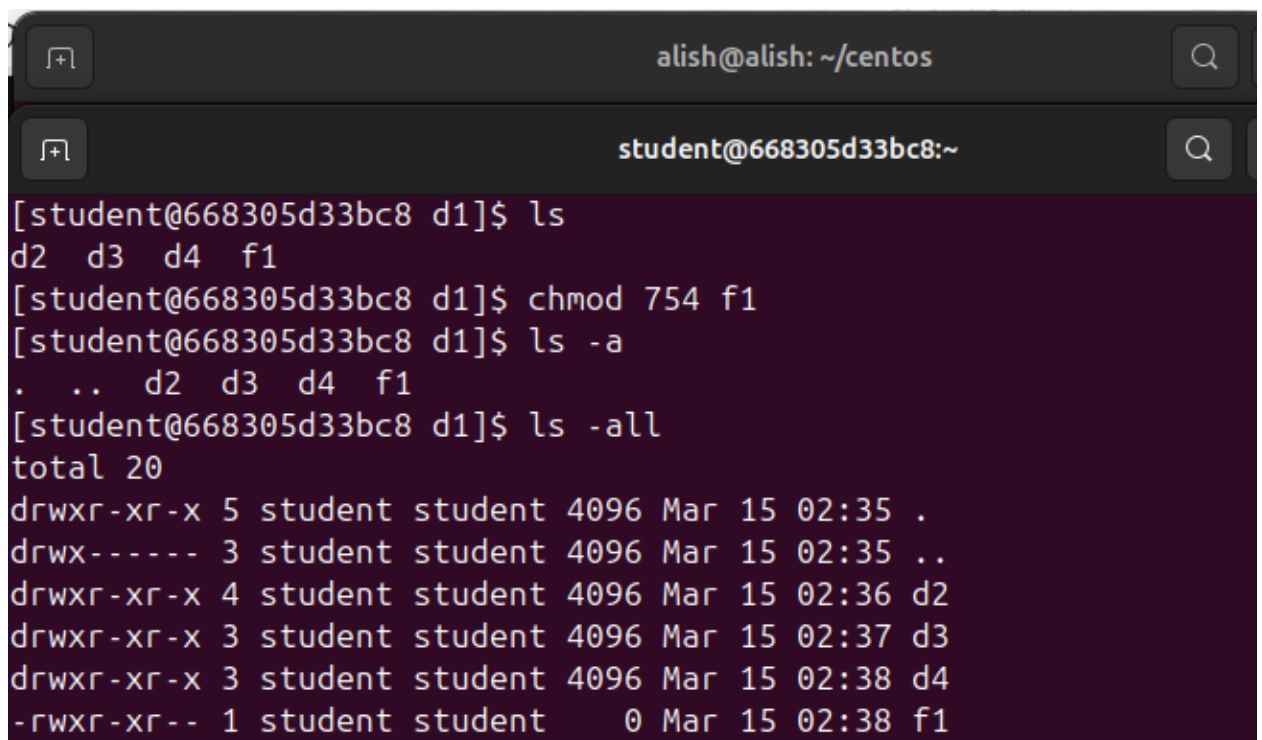


```
alish@alish: ~/centos
student@668305d33bc8:~
Last login: Sun Mar 15 02:29:12 UTC 2026 on pts/1
[student@668305d33bc8 ~]$ ls
d1
[student@668305d33bc8 ~]$ tree
.
├── d1
│   ├── d2
│   │   ├── d5
│   │   │   ├── f2
│   │   │   └── f3
│   │   └── d6
│   │       ├── f4
│   │       └── f5
│   ├── d3
│   │   └── d8
│   ├── d4
│   │   └── d7
│   │       └── f6
│   └── f1
└──
```

8 directories, 6 files
[student@668305d33bc8 ~]\$

C. Change the permission of file f1 so that the owner will get full permission, group member will get read and execute permission and other will get read permission

Command: - chmod 754 /d1/f1



```
alish@alish: ~/centos
student@668305d33bc8:~
[student@668305d33bc8 d1]$ ls
d2 d3 d4 f1
[student@668305d33bc8 d1]$ chmod 754 f1
[student@668305d33bc8 d1]$ ls -a
. .. d2 d3 d4 f1
[student@668305d33bc8 d1]$ ls -all
total 20
drwxr-xr-x 5 student student 4096 Mar 15 02:35 .
drwx----- 3 student student 4096 Mar 15 02:35 ..
drwxr-xr-x 4 student student 4096 Mar 15 02:36 d2
drwxr-xr-x 3 student student 4096 Mar 15 02:37 d3
drwxr-xr-x 3 student student 4096 Mar 15 02:38 d4
-rwxr-xr-- 1 student student   0 Mar 15 02:38 f1
```

D. Change permission of file f2 such that owner and group member will get read write permission but other will get no permission

Command: – `chmod 660 /d1/d2/d5/f2`

```
alish@alish: ~/centos
student@668305d33bc8:~
[student@668305d33bc8 d1]$ ls
d2 d3 d4 f1
[student@668305d33bc8 d1]$ cd d2
[student@668305d33bc8 d2]$ ls
d5 d6
[student@668305d33bc8 d2]$ cd d5
[student@668305d33bc8 d5]$ ls
f2 f3
[student@668305d33bc8 d5]$ chmod 660 f2
[student@668305d33bc8 d5]$ ls -all
total 8
drwxr-xr-x 2 student student 4096 Mar 15 02:36 .
drwxr-xr-x 4 student student 4096 Mar 15 02:36 ..
-rw-rw---- 1 student student   0 Mar 15 02:36 f2
-rw-r--r-- 1 student student   0 Mar 15 02:36 f3
```

E. Change permission of directory d3 such that all categories (read,write,execute) is done

Command – `chmod 777 ./d1/d3`

```
alish@alish: ~/centos
student@668305d33bc8:~
[student@668305d33bc8 d3]$ cd .
[student@668305d33bc8 d3]$ cd ..
[student@668305d33bc8 d1]$ chmod 777 d3
[student@668305d33bc8 d1]$ ls -all
total 20
drwxr-xr-x 5 student student 4096 Mar 15 02:35 .
drwx----- 3 student student 4096 Mar 15 02:35 ..
drwxr-xr-x 4 student student 4096 Mar 15 02:36 d2
drwxrwxrwx 3 student student 4096 Mar 15 02:37 d3
drwxr-xr-x 3 student student 4096 Mar 15 02:38 d4
-rwxr-xr-- 1 student student   0 Mar 15 02:38 f1
```